

Assignment 02

Linux Commands

Submitted By: M M Ahsan Uddin
Email: ahsanuddin116@gmail.com

System Update

Command	Description	Example
sudo apt update	Updates the package list from repositories.	sudo apt update
sudo apt upgrade	Upgrades installed packages to the latest versions	sudo apt upgrade

Navigation & File Management

Command	Description	Example
pwd	Displays the current working directory	pwd
ls	Lists files and directories	ls
ls -la	Lists all files, including hidden files, with detailed information	ls -la
cd	Changes the current directory	cd /var/www/htm
cp	Copies files or directories	cp file.txt /tmp/
mv	Moves or renames files/directories	mv old.txt new.txt
rm	Removes files or directories.	rm file.txt
rm -rf	Forcefully removes a directory and its contents.	rm -rf test
cat	Displays file contents.	cat /etc/passwd
nano	Opens a file in Nano text editor.	nano config.txt
vim	Opens a file in Vim editor.	vim config.txt
touch	Creates an empty file	touch hello.txt
mkdir	Creates a new directory.	mkdir myfolder

User & Permission Management

Command	Description	Example
whoami	Displays the current username.	whoami
id	Shows user and group IDs	id
sudo	Executes commands with administrator privileges.	sudo apt update
chmod	Changes file permissions.	chmod 755 script.sh
chmod +x	Makes a file executable.	chmod +x script.sh
passwd	Changes a user's password	passwd user

Networking Commands

Command	Description	Example
ping	Tests network connectivity	ping google.com
ifconfig	Displays network interface information	ifconfig
ip a	Shows IP addresses and interfaces.	ip a
netstat	Displays network connections and ports	netstat
ss -tuln	Shows listening TCP/UDP ports.	ss -tuln
nmap	Scans hosts and open ports.	nmap -sV 192.168.1.1
curl	Transfers data from or to a server.	curl http://example.com
wget	Downloads files from the web	wget https://example.com/file.zip
traceroute	Traces the route packets take to a host.	traceroute google.com
dig	Performs DNS lookups.	dig example.com
nslookup	Queries DNS records.	nslookup example.com

File Search & Manipulation

Command	Description	Example
find	Searches for files and directories	find / -name "*.php"
grep	Searches text inside files.	grep "admin" config.php
locate	Quickly finds files using a database.	locate php.ini
file	Identifies the file type.	`file shell
history	Displays previously executed commands.	history

Privilege Escalation & Enumeration

Command	Description	Example
sudo -l	Lists commands the user can run with sudo.	sudo -l
ps aux	Shows running processes.	ps aux
uname -a	Displays kernel and system information.	uname -a
env	Displays environment variables.	env

Security & Penetration Testing Tools

Command	Description	Example
msfconsole	Starts the Metasploit Framework.	msfconsole
hydra`	Password brute-force tool.	hydra -l admin -P passwords.txt 192.168.1.5 ssh
john	Password cracking tool (John the Ripper).	john --wordlist=rockyou.txt hashes.txt
nikto	Web server vulnerability scanner.	nikto -h http://target
sqlmap	Automates SQL injection testing.	sqlmap -u "http://site.com?id=1" --dbs
burpsuite	GUI web security testing proxy.	burpsuite
dirb	Directory brute-force tool.	dirb http://target
gobuster	Fast directory and DNS brute-forcing tool.	gobuster dir -u http://target -w wordlist.txt
wireshark	GUI network packet analyzer.	wireshark